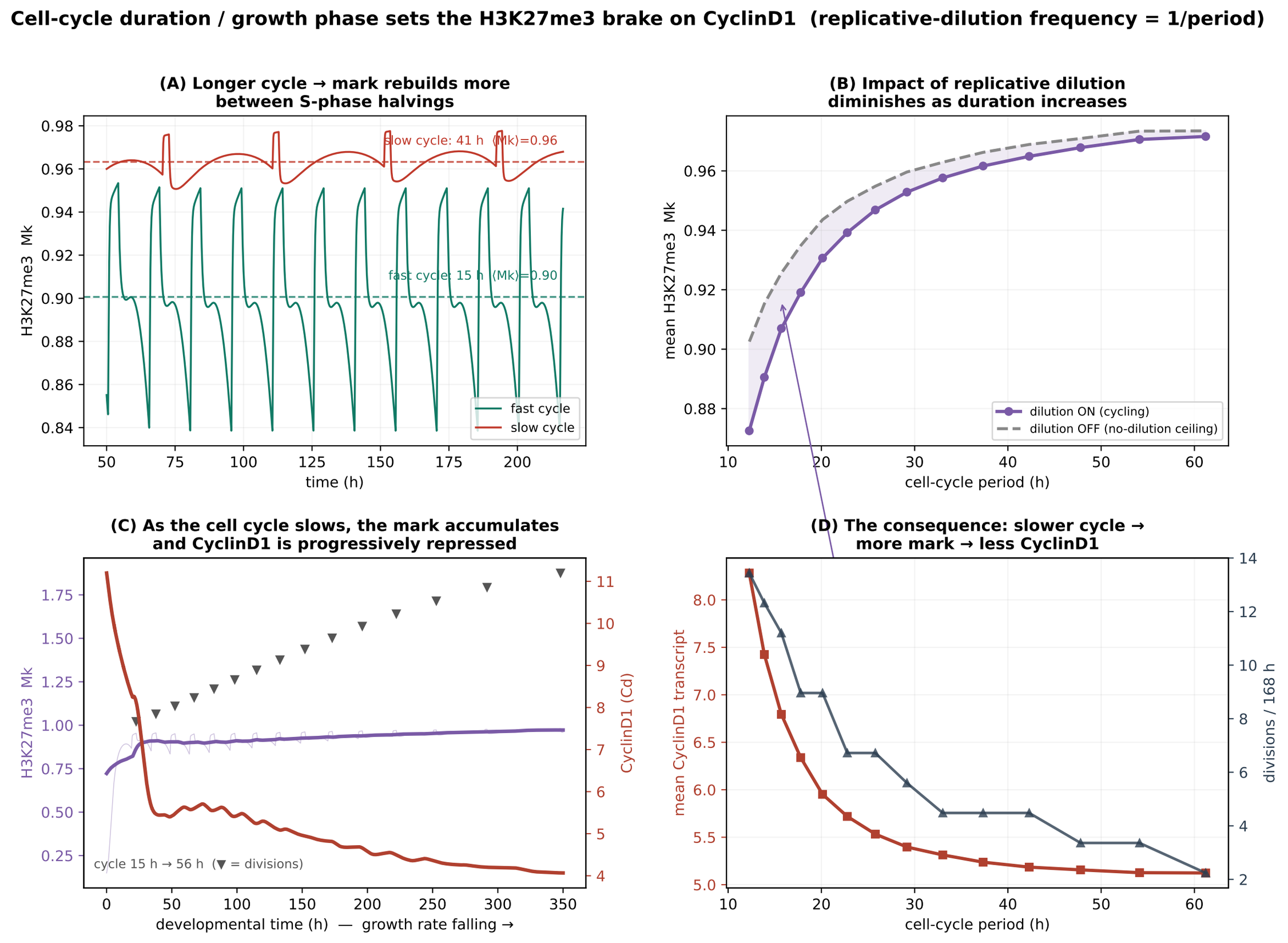
Cell-cycle duration / growth phase sets the H3K27me3 brake on CyclinD1 (replicative-dilution frequency = 1/period)
(A) Longer cycle → mark rebuilds more between S-phase halvings
slow cycle: 41 h ⟨Mk⟩=0.96
fast cycle: 15 h ⟨Mk⟩=0.90
fast cycle
slow cycle
H3K27me3 Mk
time (h)
(B) Impact of replicative dilution diminishes as duration increases
dilution ON (cycling)
dilution OFF (no-dilution ceiling)
mean H3K27me3 Mk
cell-cycle period (h)
(C) As the cell cycle slows, the mark accumulates and CyclinD1 is progressively repressed
cycle 15 h → 56 h (▼ = divisions)
H3K27me3 Mk
CyclinD1 (Cd)
developmental time (h) — growth rate falling →
(D) The consequence: slower cycle → more mark → less CyclinD1
mean CyclinD1 transcript
divisions / 168 h
cell-cycle period (h)